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CE F231
Fluid Mechanics

Sayantan Chakraborty, Ph.D., AM ASCE, MIE
sayantan.chakraborty@pilani.bits-pilani.ac.in

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Lecture 11: Buoyancy and Floatation

Learning Outcome

- Learn about buoyancy and Archimedes principle
- Stability of submerged bodies
- Centre of buoyancy (CB) and centre of gravity (CG)

https://www.bdcnetwork.com/underwater-restaurant-open-maldives

https://edition.cnn.com/travel/article/hurawalhi-5-8-underwater-restaurant/index.html

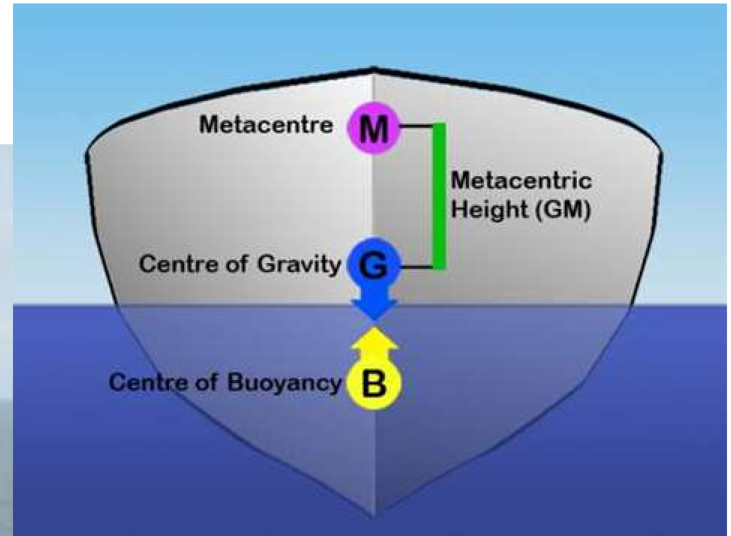
5.8 Undersea Restaurant

Learning Outcome

- Stability of floating bodies
- CB, CG and metacentre

Appomattox, a deep-water oil and gas project platform, is Shell's largest floating platform in the Gulf of Mexico.

https://www.youtube.com/watch?v=iI_N5ziD7jg&t=202s

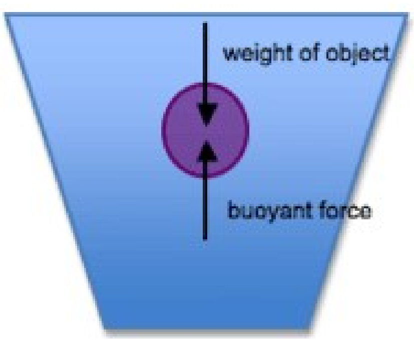


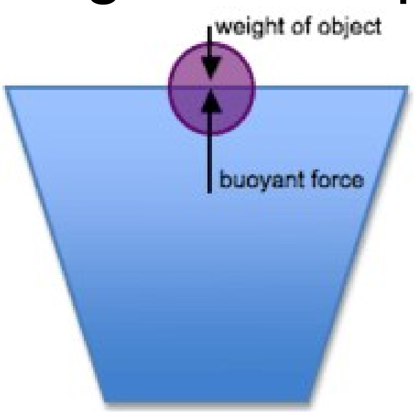
https://quizlet.com/gb/445777764/ship-stability-lo2-flash-cards/

https://lshipdesign.blogspot.com/2016/09/strength-and-stability-of-ship-or-any.html

Buoyancy and Archimedes Principle

- Archimedes principle states that any body completely or partially submerged in a fluid at rest is acted upon by an upward, or buoyant force, the magnitude of which is equal to the weight of the fluid displaced by the body.
- The volume of displaced fluid is equal to the volume of an object fully immersed in a fluid or to that fraction of the volume below the surface for an object partially submerged in a liquid.

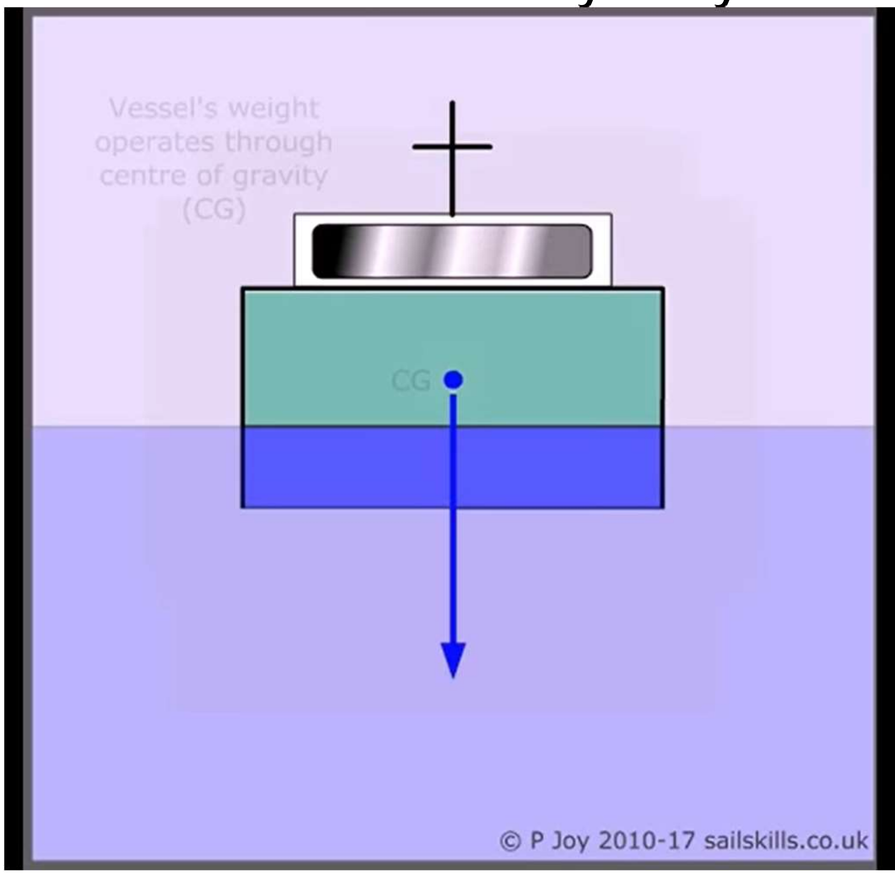




https://www.mecholic.com/2018/06/differentiate-centre-of-gravity-metacentre.html

Centre of Gravity and Centre of Buoyancy

- Centre of gravity: Point through which weight of body is supposed to act
- Centre of buoyancy: Point through which buoyant force is supposed to act. Centre of buoyancy = centroid of volume of displaced fluid



https://www.youtube.com/watch?v=a79yUWofnSY

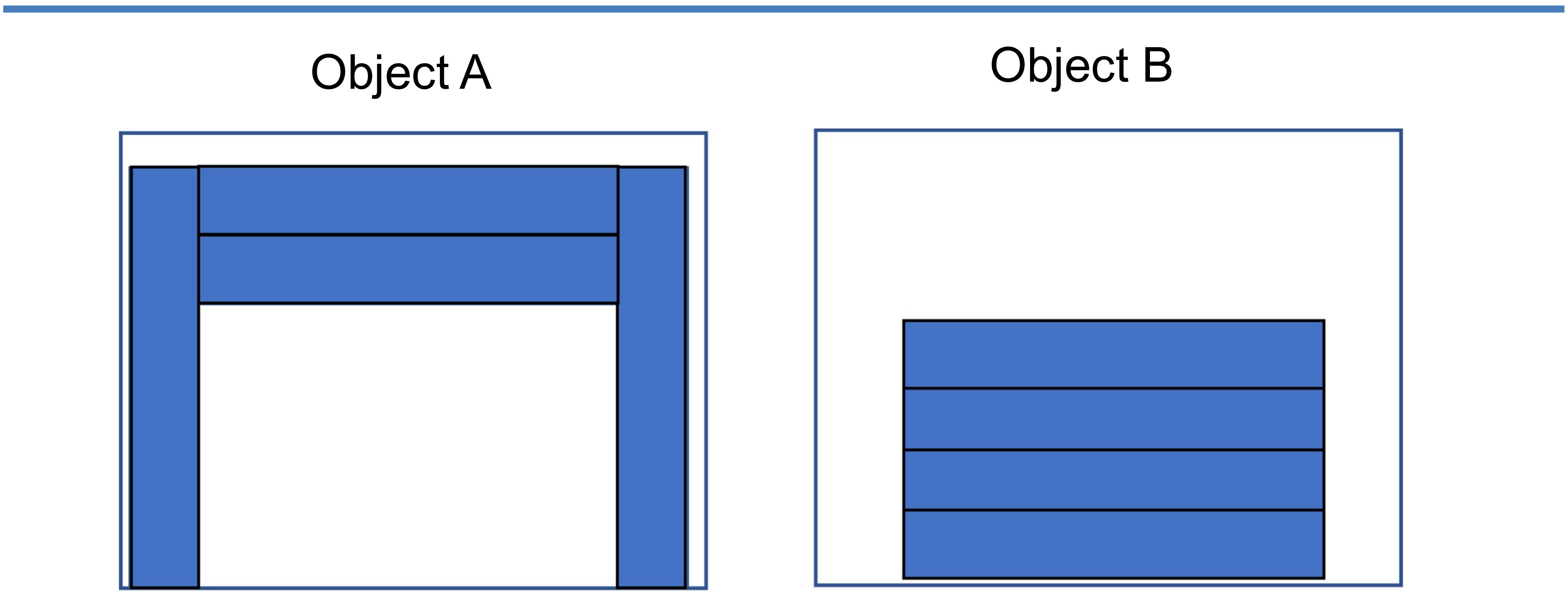
https://www.youtube.com/watch?v=AC9EULljVbo

Centre of Gravity and Centre of Buoyancy

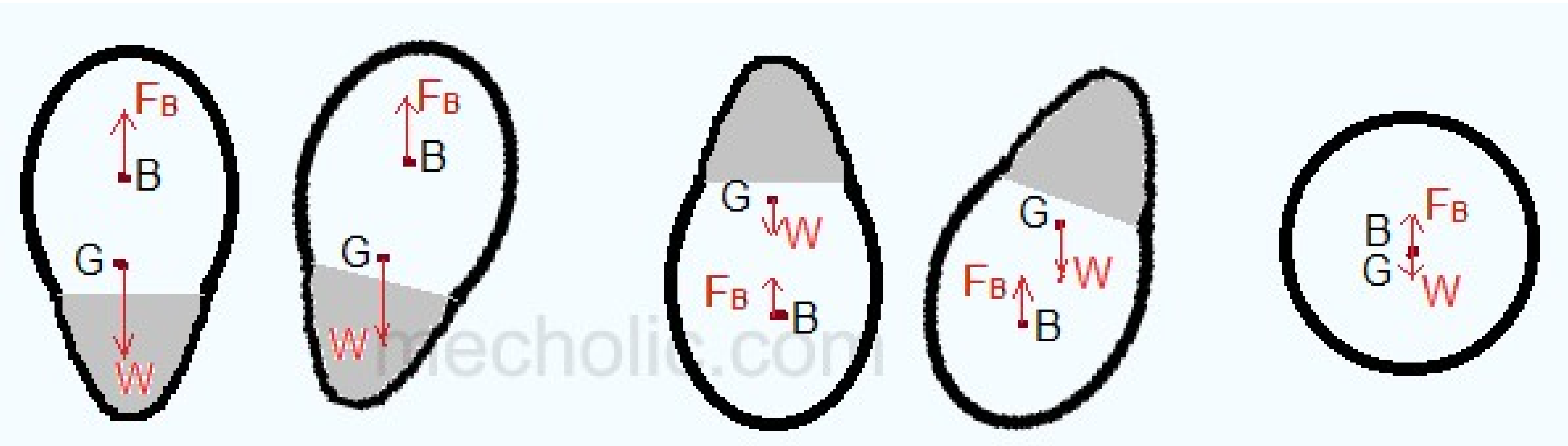
Objects A and B are completely submerged in water

a) Comment on the relative locations of CG of objects A and B

b) Comment on the relative locations of CB of objects A and B



Stability of Submerged Bodies

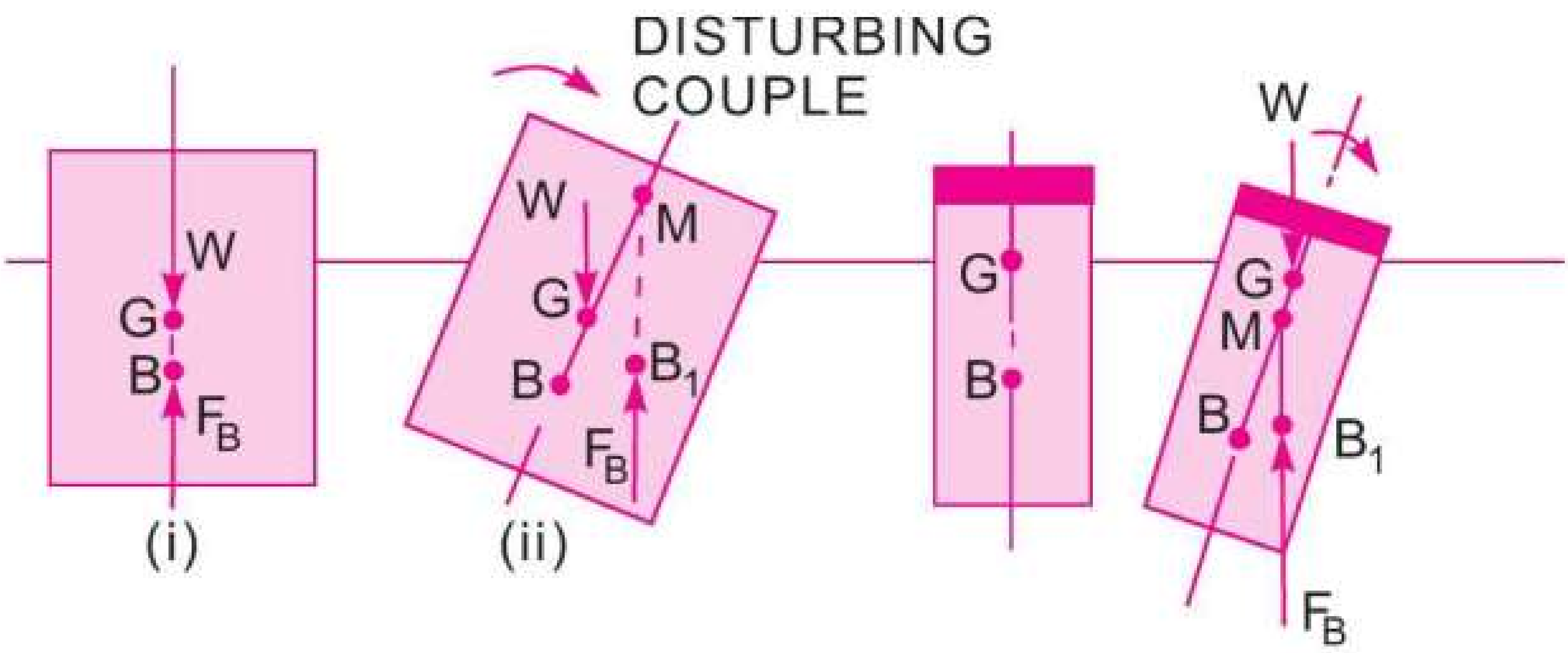


- Stable if G is below B
- Unstable if G is above B
- Neutral if G and B coincides

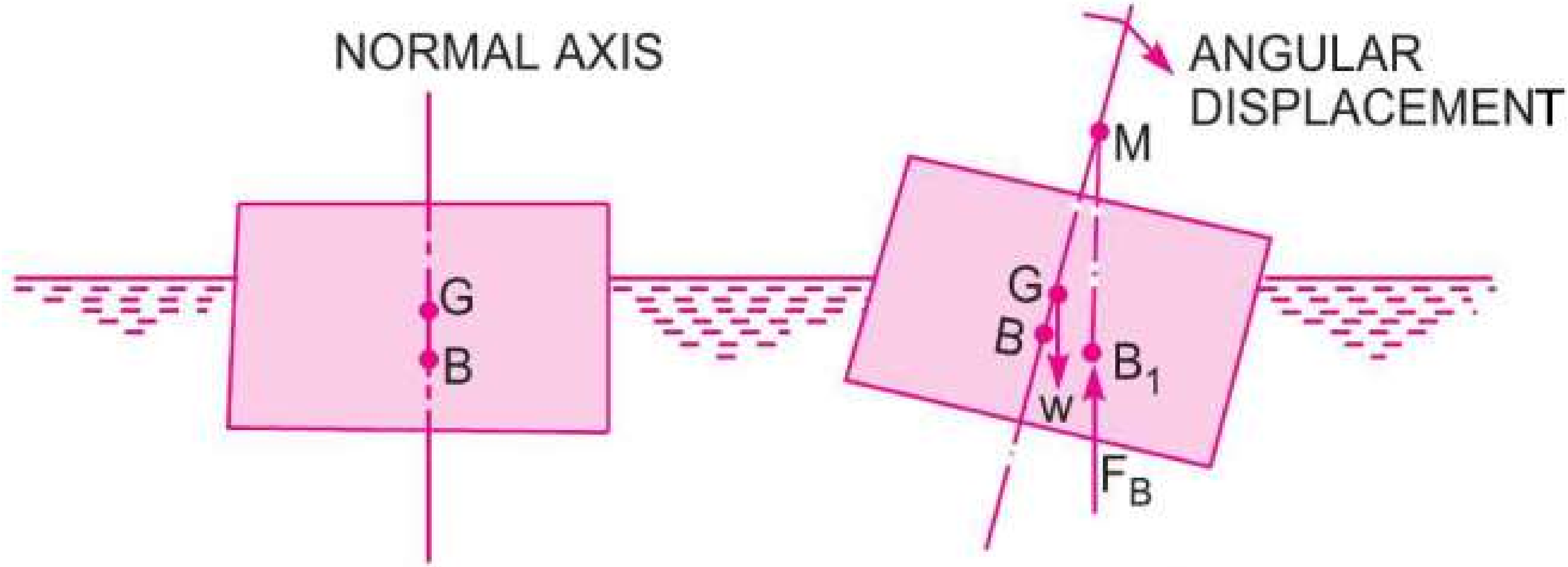
Stable or unstable → classified based on whether the body is able or unable to regain initial position after disturbing/tilting

Stability of Floating Bodies

- Unlike submerged bodies, we cannot comment on stability of floating bodies solely based on relative locations of G and B (i.e., based on G below or above B)



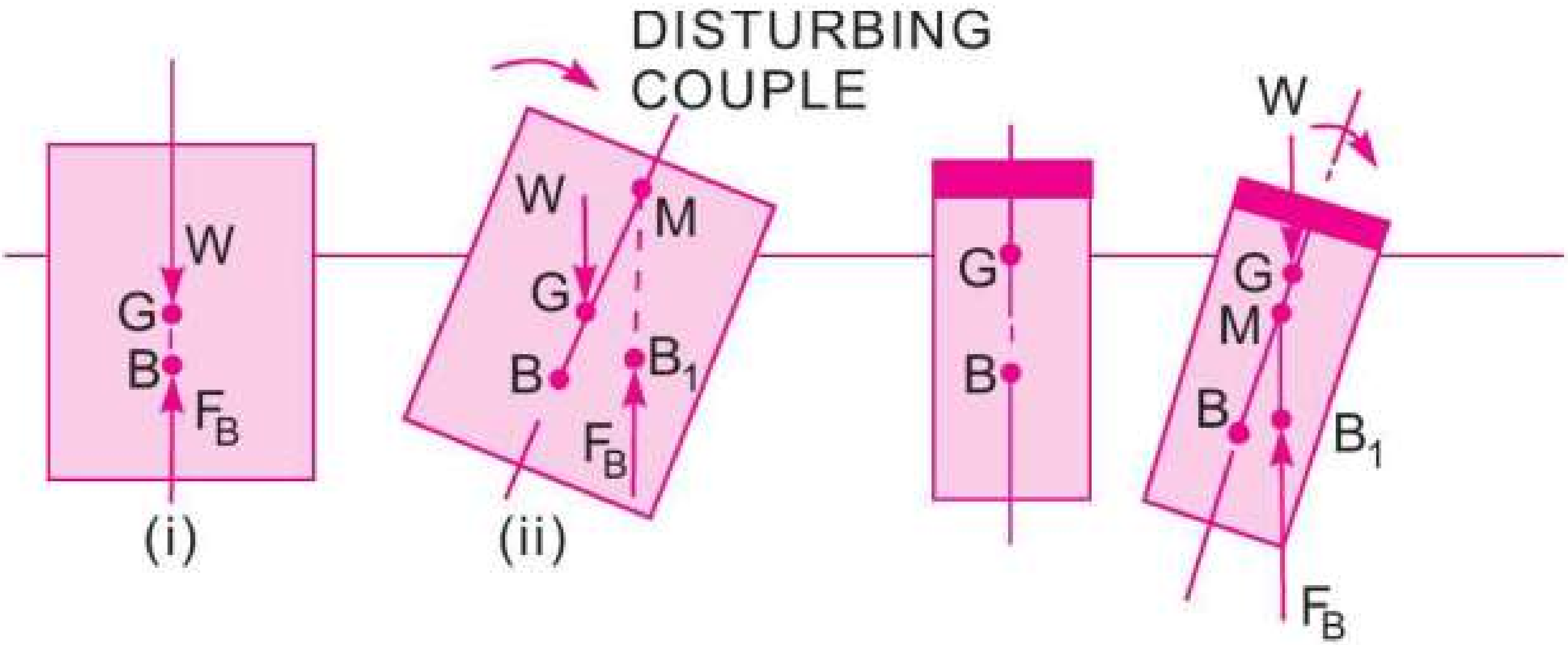
Metacentre



(a) (b)

- After imparting a small angular displacement, the line of action of buoyant force will intersect the original normal axis at point M – metacentre

Metacentric Height and Stability of Floating Bodies



- Metacentric height GM = distance b/w CG and M
- GM = BM – BG
  - GM → +ve → M is above G → stable
  - GM → -ve → M is below G → unstable
  - GM → 0 → G and M coincides → neutral